Agentic Generative AI for Clinical Decision Making in Treatment, Payment, and Operations

Chain Reasoning, Tool-Calling, and Retrieval-Augmented Agents as the Orchestration Layer for Payment Integrity

Thesis: *Payment integrity is shifting from single-shot models that score a claim to agentic generative AI that reasons through it. These systems plan, work step by step, pull in the governing policy as evidence, and call specialized models as tools. For Cotiviti, the opportunity is clear: deploy reasoning agents as an orchestration layer over its proven pattern-recognition models, turning fast flags into cited, auditable decisions.*

1. The Problem in Healthcare TPO

A suspicious claim is easy to spot and hard to resolve. Cotiviti's models can flag one almost instantly, but a human auditor may still spend significant time proving why it should be denied, citing the exact policy that makes the case. Multiply that by millions of claims and the bottleneck becomes clear. Every claim is a decision that spans Treatment, Payment, and Operations (TPO): Is the service medically necessary? Does it follow payer policy and correct-coding rules? Does the billing look like fraud, waste, or abuse (FWA)? Cotiviti answers these at national scale, eliminating over $9.5 billion in payment waste in 2024 (Cotiviti, 2025). Narrow machine-learning models do the first pass: classification sorts codes, prediction scores risk, clustering groups provider behavior, and anomaly detection flags billing that drifts from the norm. Human auditors resolve what they surface.

Two pressures are breaking this balance. Regulatory change keeps inventing new billing schemes, so payers must watch utilization patterns to catch "creative billing" the moment it appears (Cotiviti, 2026a). At the same time, providers now use AI to optimize coding and fire off appeals at machine speed (Cotiviti, 2026b). The needle is no longer the problem; the models find it fast. The problem is explaining why it is a needle, against the right policy, faster than the appeals pile up. That is the gap agentic generative AI is built to close.

Agentic generative AI adds four capabilities to a large language model (LLM). Chain-of-thought reasoning breaks a medical-necessity check into explicit steps. Tool-calling lets the agent invoke Cotiviti's existing classification, clustering, and anomaly models as functions rather than replace them. Retrieval-augmented generation (RAG) pulls the governing policy from a vector store to ground its reasoning in citable evidence. Memory and orchestration let specialized agents coordinate on a case and escalate anything ambiguous to a human. The flow is simple: take in a claim, retrieve its policy, reason over the pair, call the risk models, and return a recommendation with a provenance trail. The difference is an explainable determination instead of an opaque score.

2. Relevant Trends

From single models to teams of agents. The industry is moving away from one big model that does everything. Gartner forecasts that U.S. health plans will increasingly deploy an "agentic workforce" in 2026 to automate prior authorization, claims processing, and payment integrity, as long as governance and human oversight keep pace (Cohere Health, 2026). The shift is architectural: vendors now coordinate teams of specialized agents that hand work to one another, rather than relying on a single monolithic model (Vantage Point, 2026), and they continuously retune those models as payer policies change (Innovaccer, 2026).

It is already shipping. This is not a forecast waiting to happen. CVS Health and Aetna have deployed a second-generation agentic Claims Assist Manager to speed claims processing and improve payment accuracy (CVS Health, 2026). Closer to home, Cotiviti's own Edifecs business runs task-oriented agents for value-based payment workflows (Edifecs, 2025). Research is keeping step: a 2026 IEEE taxonomy sets out how to evaluate healthcare AI agents across seven dimensions before they go live (Vatsal et al., 2026), and retrieval grounding plus structured outputs have become the go-to techniques for keeping these agents from hallucinating in clinical settings.

The provider-side catalyst. The biggest pressure comes from the other side of the claim. Providers now use AI to optimize coding and auto-generate appeals, which Cotiviti calls a present-day reality reshaping how claims are submitted (Cotiviti, 2026b). When the other side can generate optimized claims and appeals at machine scale, payers need agents of their own just to respond at the same speed.

3. Opportunities and Threats

Opportunities:

- **Defensible decisions, not just flags.** RAG-grounded agents cite the exact policy language and model signal behind every recommendation. In a market where providers appeal aggressively, an explanation a reviewer can stand behind is worth as much as the catch itself.
- **Build on what already works.** Tool-calling lets agents use Cotiviti's mature classification, clustering, and anomaly models as callable functions. This extends proven IP rather than rebuilding it, which means faster payoff and lower risk.
- **Catch it before the money goes out.** Pairing agentic reasoning with anomaly detection moves review ahead of payment, cutting the cost and friction of chasing dollars back after the fact (Cotiviti, 2026a).

Threats:

- **The model invents a reason.** Ungrounded generation can fabricate a justification that looks plausible but is wrong. The defenses are RAG grounding, structured outputs, and a human in the loop on every determination (Vatsal et al., 2026).
- **No audit trail, no defense.** Autonomous decisions that affect payment or care access invite regulatory and legal challenge if the system cannot show its work (Cohere Health, 2026).
- **An AI arms race.** When provider and payer agents generate claims and appeals against each other, review pipelines can flood faster than people can clear them (Cotiviti, 2026b).
- **Everyone is shipping this.** Automation and EHR-adjacent vendors are already releasing competing agentic products, so the window to differentiate is narrowing (UiPath, 2026; Innovaccer, 2026).

4. Strategic Recommendations for Cotiviti

These recommendations are built to make the clinical chart validation team's existing work faster and better, not to replace it. Cotiviti's auditors already do the hard part: nurses and coders pull a flagged DRG claim, then check whether the billed diagnoses, codes, and modifiers are supported by the medical record and the governing guideline (Cotiviti, 2020). The goal is to put an agent alongside them that does the gathering and drafting, so the human spends time on judgment, not assembly.

- **Give each auditor a "first-draft" agent for flagged DRG claims.** Only 1–3% of claims reach clinical review, so each one counts (Cotiviti, 2020). When the analytics select a chart, a retrieval-augmented agent pulls the relevant national and proprietary coding guidelines, lines up the billed codes against the documented clinical indicators, and hands the auditor a draft rationale with citations. The nurse still decides; the agent removes the manual hunt through guidelines and records, attacking the 90-day time-to-value problem in DRG review (Cotiviti, 2024b).
- **Target the modifier-justification problem first.** Cotiviti's own audits found roughly 38% of overriding modifiers were unjustified, mirroring the OIG's ~40% finding on modifier 59 (Cotiviti, 2020). Point the agent's first use case here: flag and explain likely modifier misuse against policy, giving coders a head start on a high-yield error type with a known baseline to beat.
- **Bake provider-contract rules into the agent.** Auditors must honor provider-specific constraints, such as a provider excluded from Medicaid audits or one auditable only within 120 days of payment (Cotiviti, 2020). Encode these as guardrails the agent checks automatically before surfacing a claim, removing a manual, error-prone step and protecting provider relations.
- **Keep the human in the loop and show the work**. Every agent output carries a plain-language rationale linked to the guideline text and record evidence behind it. This fits Cotiviti's "augmented intelligence" philosophy and its 2015 lesson that AI underperforms when cut off from domain expertise (Cotiviti, 2024a), while giving the team an audit trail that strengthens appeals defense.

Together, these moves speed the work the chart validation team already does: less time assembling evidence, more time on clinical judgment, with a citation trail that holds up on appeal.

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